

Vinayak Paroonon Kooloth

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Mobile: +49 15560026916

Location: Oulu, Finland

Open to relocation across EU

EDUCATION

- **Hof University of Applied Sciences** Hof, Germany
Master of Science in Artificial Intelligence & Robotics Mar. 2024
- **CMS College of Engineering and Technology** India
Bachelor of Engineering in Computer Science and Engineering Aug. 2019 – Jul. 2023

EXPERIENCE

- **Monidor Oy** Oulu, Finland
AI & Full-Stack Engineer – Medical Systems Jan. 2026 – Present
 - **Patient Vitals Reporting System (Node.js):** Built a Node.js backend that queries and joins time-series physiological data (SpO₂, EtCO₂, capnography waveforms) from PostgreSQL, generating server-rendered HTML reports accessible in-browser and exportable for clinical review.
 - **Clinical Data Reliability:** Engineered robust data retrieval logic to handle missing sensor readings, device error states, and high-latency DB queries, ensuring report accuracy and system reliability in a clinical environment.
 - **Multilingual AI Clinical Assistant:** Built an end-to-end clinical query system integrating a fine-tuned XLM-RoBERTa intent parser (supporting 4+ languages via domain-specific fine-tuning and prompt engineering), adaptive RAG-based retrieval from patient history, and an LLM API layer for answer synthesis – delivering clinician responses in 1–3 seconds.
 - **Multi-Strategy RAG Research & Evaluation:** Designed and benchmarked multiple retrieval strategies – standard RAG, temporal RAG, multi-hop RAG, and hybrid dense-sparse retrieval – finding that optimal strategy varied by clinical query type; implemented adaptive retrieval routing to select the best approach per query class.
 - **Legacy Backend Integration:** Refactored legacy Node.js backend and updated PostgreSQL schemas to support new physiological data types (SpO₂, EtCO₂), ensuring data persistence and compatibility across both the reporting and AI assistant systems.
 - **AI Agent usage:** Uses multiple AI agents session to help in coding e.g. Cluade code, Codex, Open Source Local model KImi 2.6 - as Tools for support only.
- **Fricke und Mallah Microwave Technology GmbH** Peine, Germany
Working Student – AI Agent Development Jun. 2025 – Sep. 2025
 - **GenAI Internal Assistant:** Led development of a LangChain- and Mistral-7B-powered assistant integrating multilingual LLM agents with the company CRM and ERP systems, streamlining project documentation and reporting workflows.
 - **RAG Pipeline for Email Automation:** Implemented FAISS-powered RAG for automated email triage and knowledge retrieval, reducing manual processing effort by over 60% and accelerating project follow-ups.
 - **Data Preparation & AI Readiness:** Designed preprocessing pipelines to clean, normalise, and vectorise project and ERP data, enabling high-quality sentence-transformer embeddings and improving downstream search and generation performance.
 - **Prompt Engineering & Evaluation:** Created and iteratively refined system and user prompts to maximise answer quality across multilingual queries, guided by human evaluation and automated testing for factuality and relevance.
 - **Model Benchmarking:** Benchmarked Mistral-7B and Llama-3 for automated reporting, multilingual summarisation, and knowledge extraction; presented findings and recommendations to management.
- **Technische Universität Chemnitz** Chemnitz, Germany
Student Assistant – Surgical Robotics Simulation Mar. 2025 – Oct. 2025
 - **Robotics Simulation:** Developed gesture-based lamp control and modular robotic simulations in ROS2, MediaPipe, OpenCV, and Gazebo, enabling intuitive human-robot collaboration for surgical settings.
 - **Robotic Arm Prototyping:** Prototyped a multi-joint robotic arm in Unity and RViz for an EU-funded research project, creating a fully functional simulation model for cross-lab experiment replication.
 - **Research Infrastructure:** Authored reproducible simulation setups and supported cross-departmental data sharing and experiment replication across multiple labs and clinical collaborators on an EU-funded project.

- **Hof University of Applied Sciences** Hof, Germany
Student Researcher – Applied Machine Learning Oct. 2024 – Mar. 2025
 - **Explainable AI Pipeline:** Developed a phishing detection system using DistilBERT and SHAP, achieving interpretable classification outputs suitable for security tooling integration.
 - **Model Deployment & Tracking:** Deployed the pipeline as a FastAPI microservice for real-time inference; implemented MLflow for full experiment reproducibility, tracking accuracy, loss, and latency across all training runs.
- **Bachelor's Thesis** India
Banana Leaf Disease Detection Using Hybrid Deep Learning Model 2023
 - **Hybrid CNN–LSTM Architecture:** Developed a CNN–LSTM model combining spatial feature extraction with temporal dependency learning, achieving 94.5% test accuracy on Sigatoka and Xanthomonas disease classification – outperforming standalone CNN baselines.
 - **Dataset & Evaluation:** Curated and augmented a dataset from the Mendeley repository; benchmarked CNN vs. CNN–LSTM using precision–recall and latency analysis across multiple training epochs.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript (Node.js), C++, SQL, Bash
- **LLM & RAG:** LangChain, FAISS, LlamaIndex, Phi-4, Mistral-7B, Llama 3, Temporal RAG, Multi-hop RAG, Hybrid Retrieval, Prompt Engineering, Vector Databases
- **Machine Learning:** TensorFlow, PyTorch, Scikit-learn, DistilBERT, LSTMs, CNNs, SHAP, MLflow
- **Backend & APIs:** FastAPI, Node.js, REST APIs, PostgreSQL, SQLite, Docker, CI/CD (GitHub Actions)
- **Data & Cloud:** AWS (S3, Lambda, SageMaker), Azure, Power BI, Sentence Transformers
- **Computer Vision:** OpenCV, MediaPipe, YOLOv7, Object Detection
- **Simulation & Robotics:** ROS2, Gazebo, Unity3D, RViz, Behavior Trees

LANGUAGES

- **English:** C1 – Professional proficiency **German:** B1 – Working proficiency **Finnish:** A1 – Beginner

INTERESTS

Agentic AI, RAG Systems, LLM Fine-tuning, Multilingual NLP, Computer Vision, Reinforcement Learning, Cloud Automation, Hiking, Science Fiction